Introduction to Supervised Learning

Question - Is it going to rain today? Do I need my umbrella?

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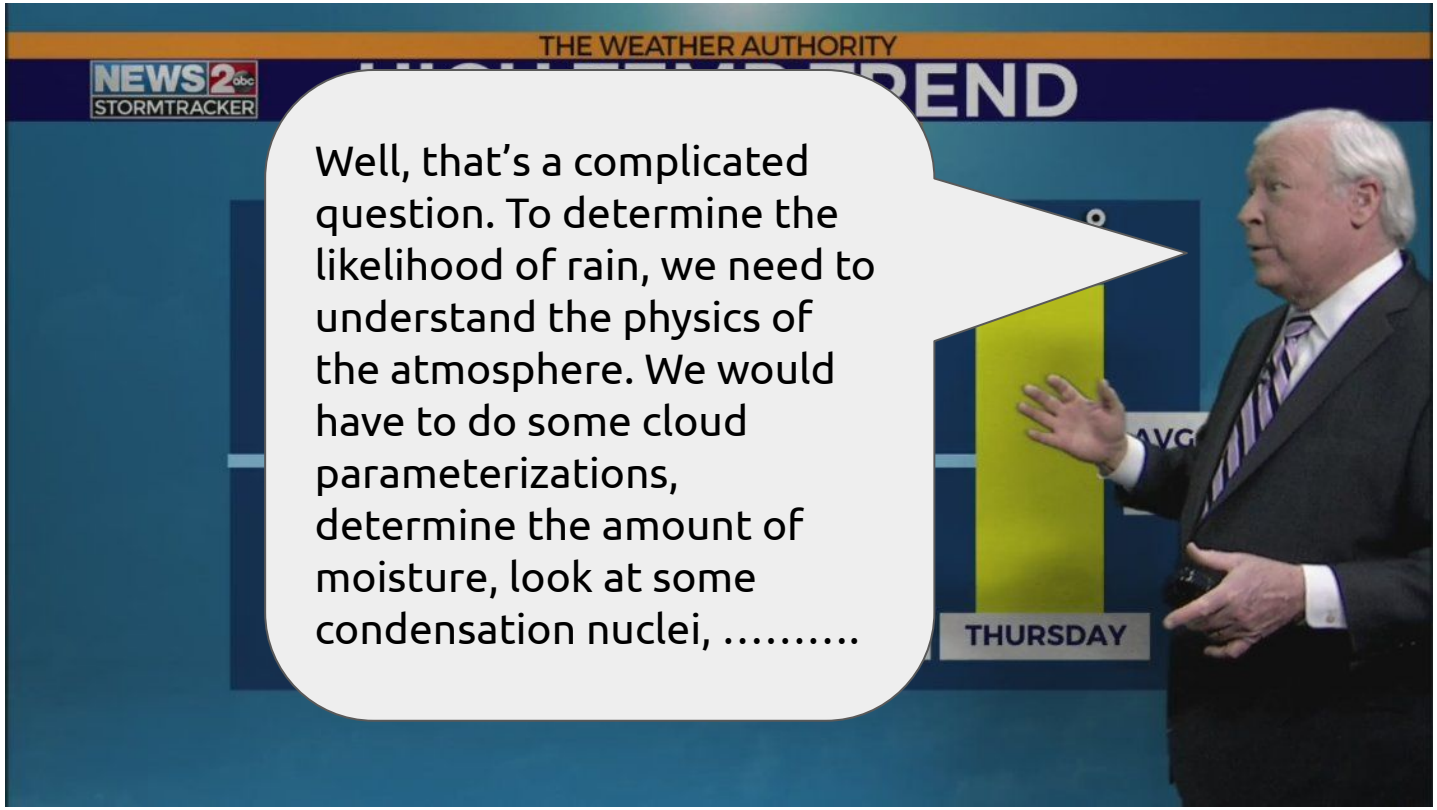

Well, that's a complicated question. To determine the likelihood of rain, we need to understand the physics of the atmosphere. We would have to do some cloud parameterizations, determine the amount of moisture, look at some condensation nuclei, …….

Question - Is it going to rain today? Do I need my umbrella?

You look outside and it looks like this:

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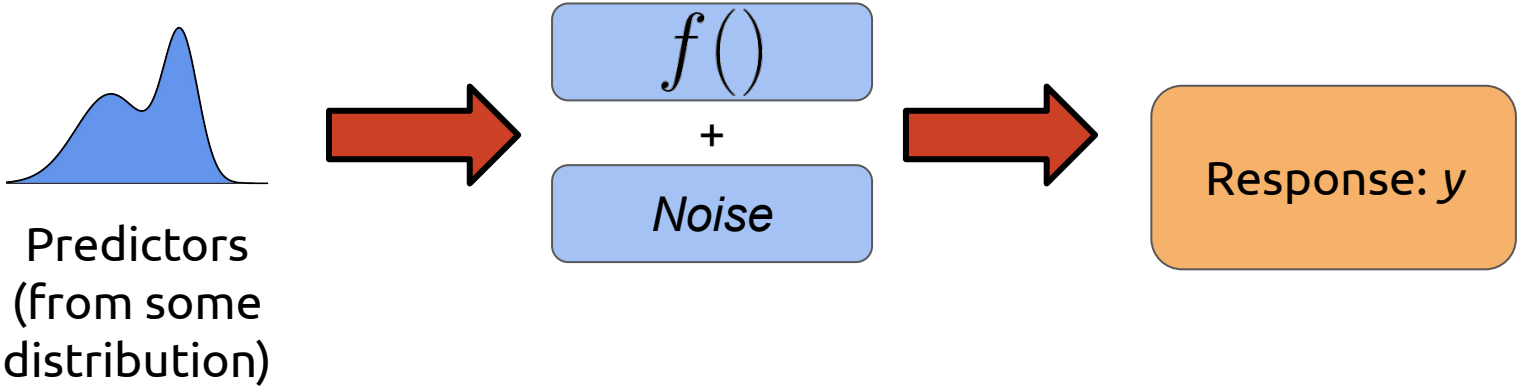
You look outside and it looks like this:

Moral: Given a good set of predictor variables and enough “experience” (training data), we can often make good predictions.

Supervised Learning - Setup



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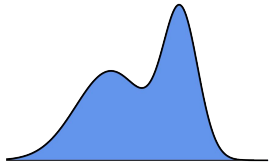


Predictors (from some distribution)

$f()$

+

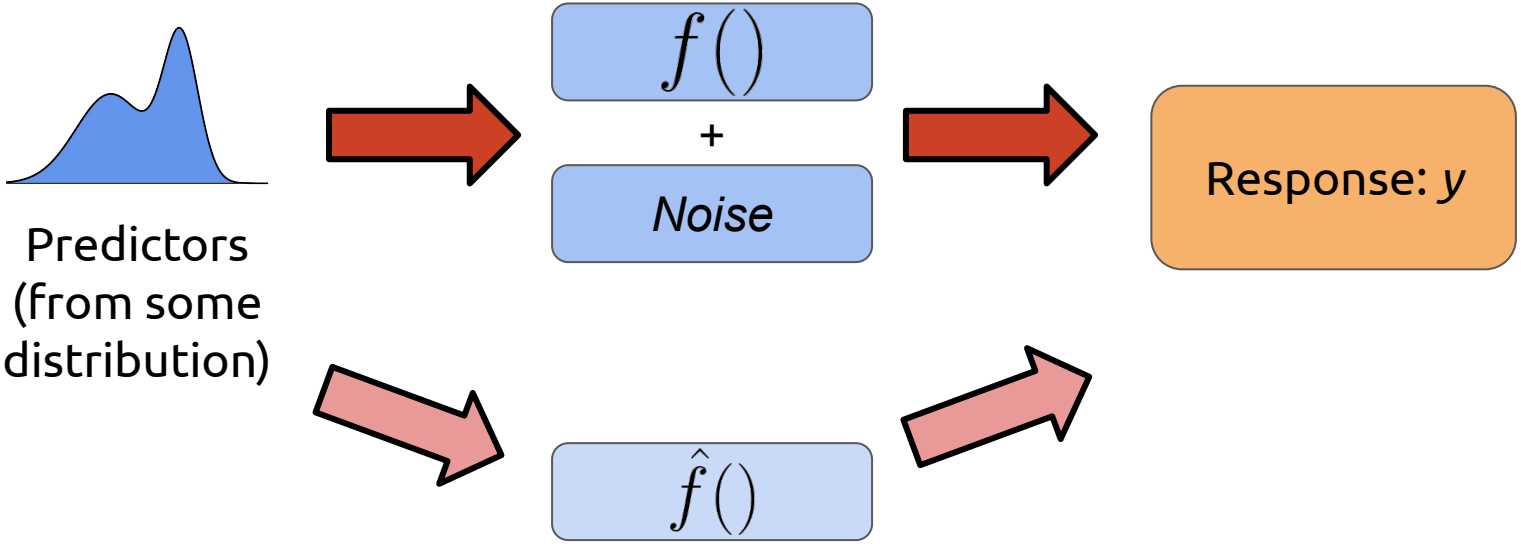
Noise

Response: *y*

Categorical: Classification

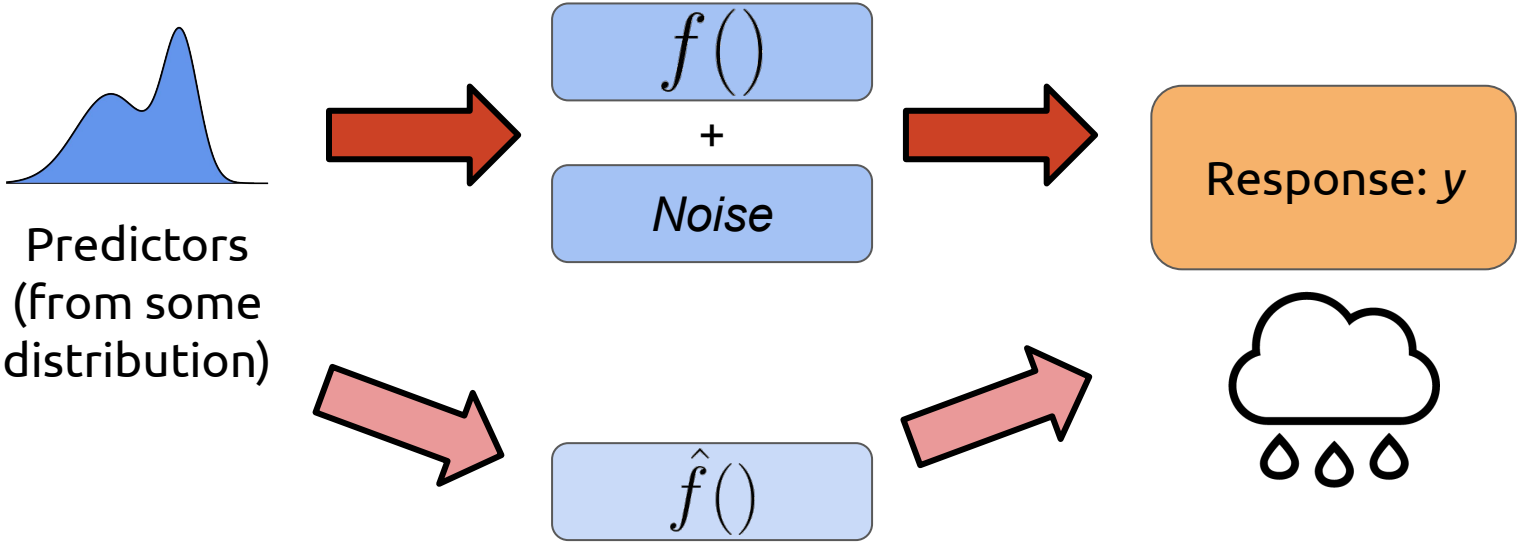
Numeric: Regression

Supervised Learning - Goals



Goal: Choose a function so that the our predictions are close (on average) to the true values.

Supervised Learning - Grossly Oversimplified



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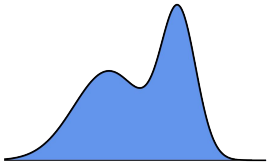

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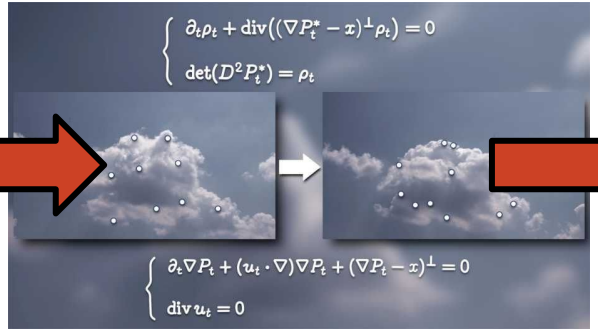
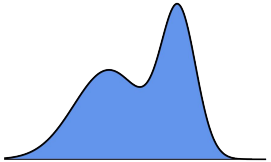


$\hat{f}()$

Response: y

Supervised Learning - Grossly Oversimplified



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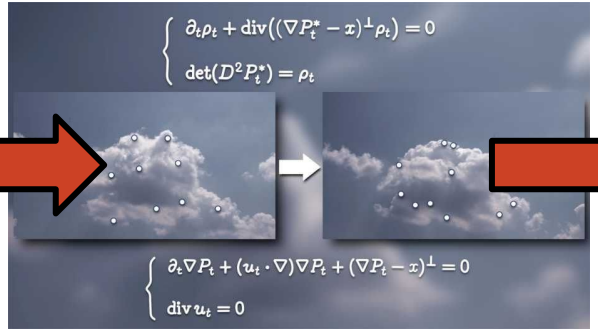
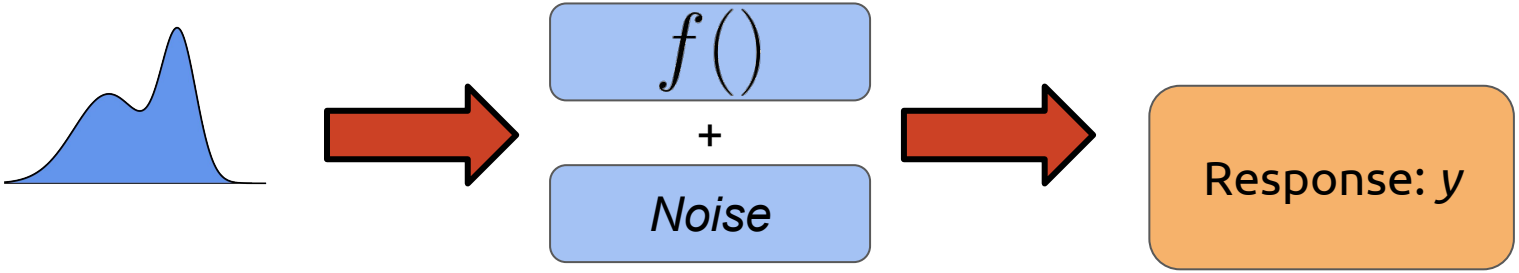


Response: *y*

My knee is acting up. Must be rain coming.

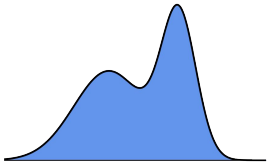
Example - Readmission or Death of Stroke Patients



90 day
readmission or
mortality

https://www.mdpi.com/2076-3417/10/18/6337/pdf

Example - Readmission or Death of Stroke Patients



demographic data
initial vital signs
laboratory results
past medical history
comorbidities
treatment-seeking behavior
pre-stroke functional status

$f()$

+

Noise

$\hat{f}()$

Response: *y*

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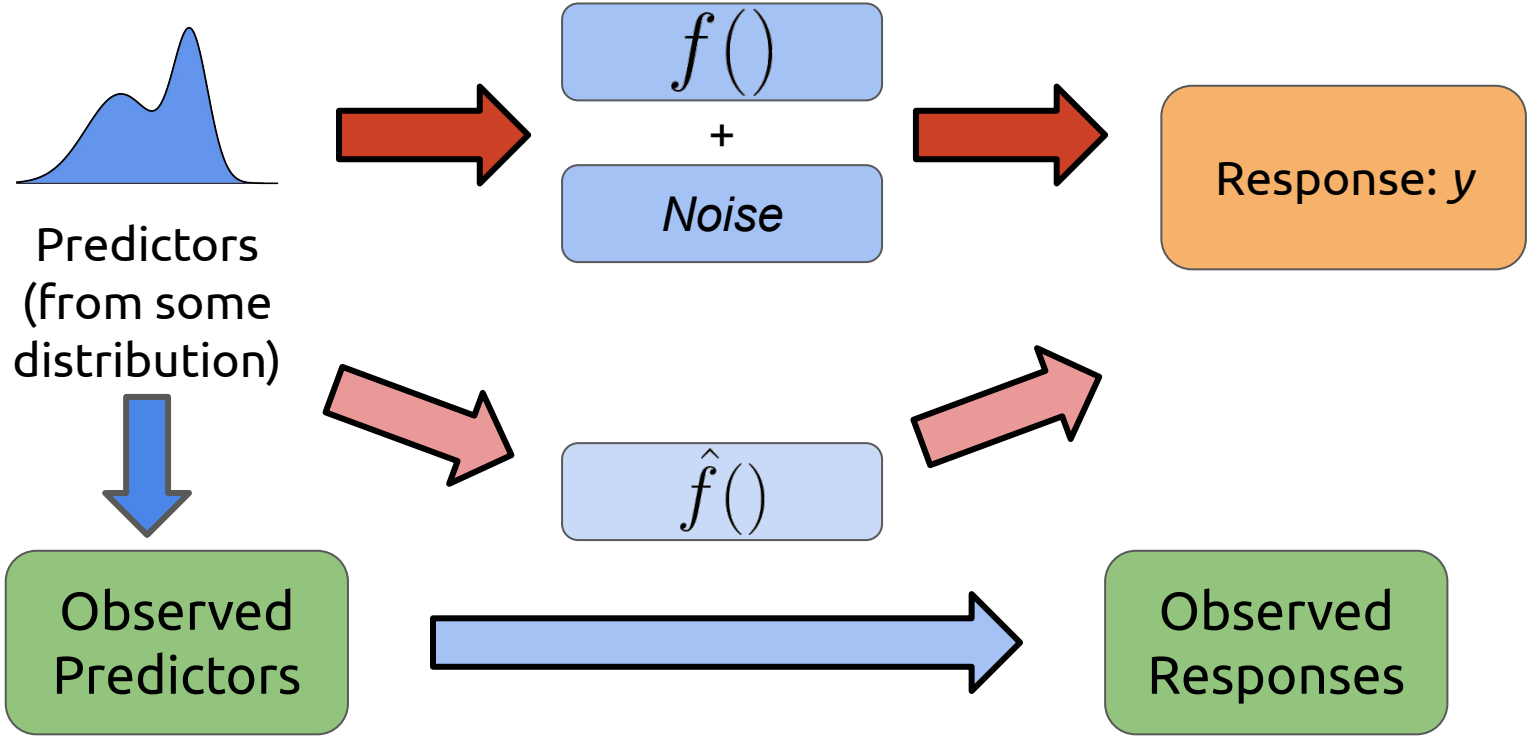
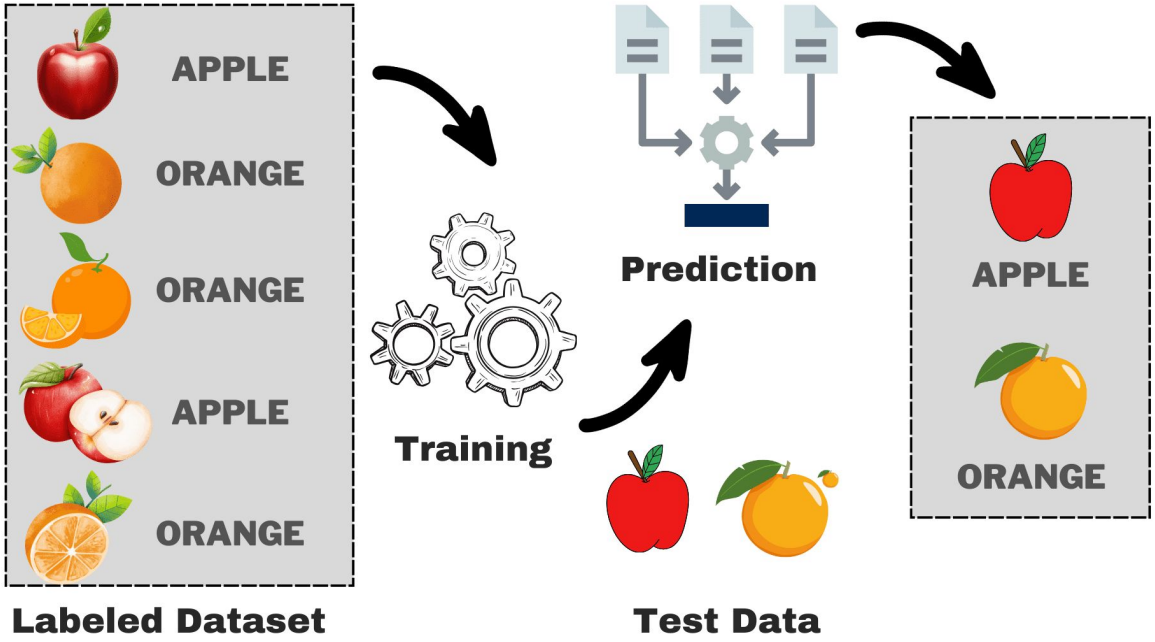
Supervised Learning - How



Supervised Learning - How

A supervised machine learning model is not give explicit instructions, but is **trained** on labeled data.

Some kind of algorithm is used to update how the model makes predictions based on the labeled examples it sees.



https://www.kdnuggets.com/understanding-supervised-learning-theory-and-overview

Supervised Learning - Goals

To measure how “good” our model is, we need some way to measure “error”.

Our goal is to minimize the expected loss over *new* data.

Important: We are not trying to minimize loss over the observed data (which is often very easy to do), but to minimize the *generalization error* - the performance on unseen data.

Measuring Generalization Data - How

If we only care about how well our model performs on unseen data, how do we measure that?

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We can’t - it’s unseen!

Measuring Generalization Data - How

If we only care about how well our model performs on unseen data, how do we measure that?

We can’t - it’s unseen!

But, we can *estimate* it.

Measuring Generalization Error - How

The most simple way to estimate generalization error is through employing a train/test split.

Full Dataset

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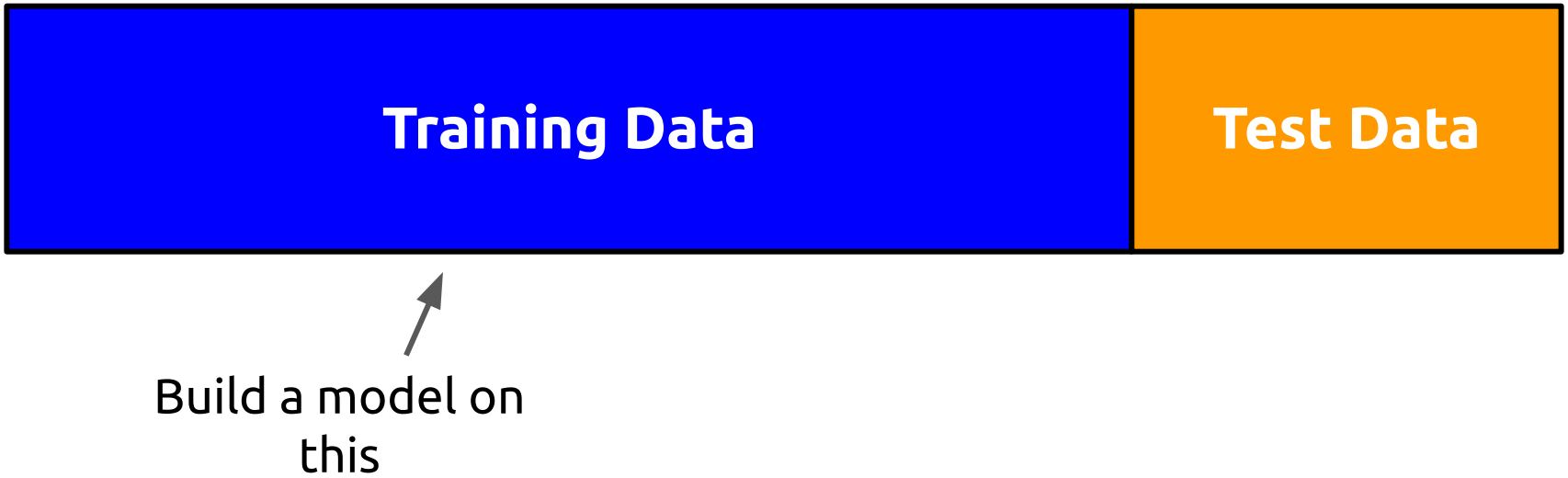

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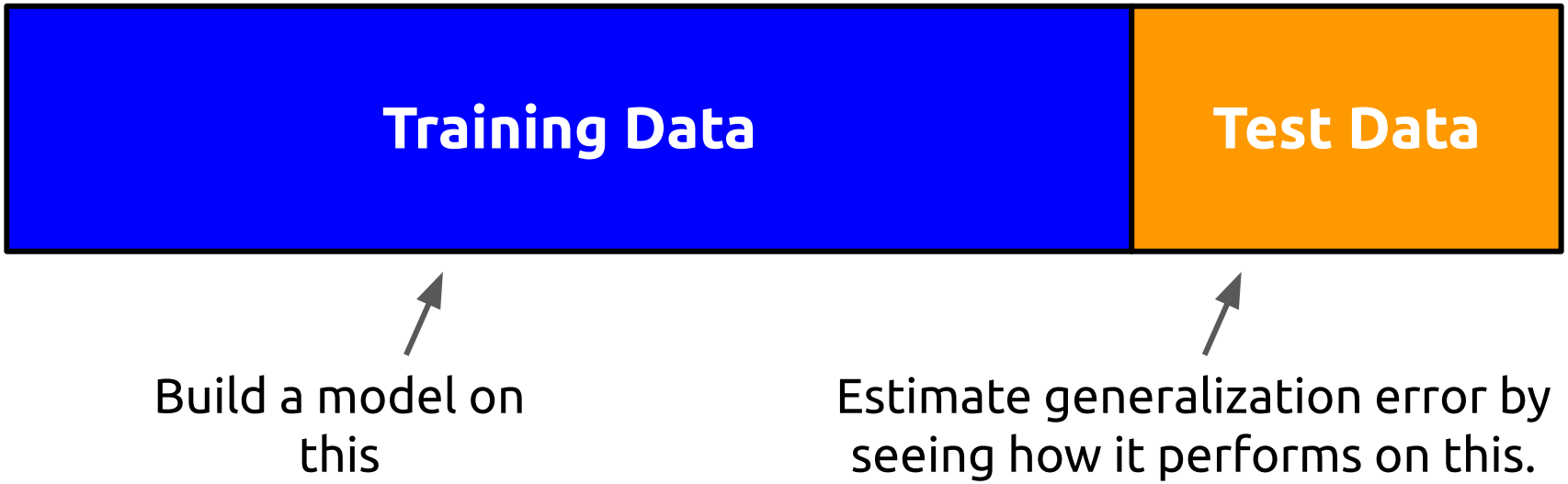

Metrics for Regression

Regression

Recall that a **regression** model tries to predict the value of a numeric target by learning patterns from labeled training data.

Examples:

- Predicting home prices
- Estimating customer spending
- Forecasting temperatures

Regression Metrics

Some common metrics for evaluating a regression model are

- (Root) Mean Squared Error (MSE/RMSE)
- Mean Absolute Error (MAE)
- Mean Absolute Percentage Error (MAPE)
- R^2 (Coefficient of Determination)

Residuals

The listed regression metrics are all based on the **residuals**, which are the differences between the actual and predicted values.

Given a dataset $\{(x_1, y_1), (x_2, y_2),\ldots,(x_n, y_n)\}$ and a predictor function $\hat{f}$ the residuals are given by:

$$\text{residual}_i = \hat{f}(x_i) - y_i$$

Mean Squared Error

$$MSE = \frac{\sum_{i=1}^{n} (\hat{f}(x_i) - y_i)^2}{n}$$

Downsides: It is measured in square units.

Eg. if predicting price, MSE will be in squared dollars.

Outlier values can be highly influential on MSE.

Root Mean Squared Error

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(\hat{f}(x_i) - y_i)^2}{n}}$$

Pros: It is measured in the same units as the target.

Cons: It is a bit difficult to directly understand what it is measuring.

Mean Absolute Error

$$MAE = \frac{\sum_{i=1}^{n} |\hat{f}(x_i) - y_i|}{n}$$

Pros: Very easy to interpret - it is exactly the average error.

Cons: Don’t have a nice way to minimize it on the training data (like we can with MSE).

Mean Absolute Percentage Error

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{\hat{f}(x_i) - y_i}{y_i}\right|$$

Pros: Normalizes the error by the size of the target. If you have a wide range of target values, this can be more informative than just calculating absolute errors.

Cons: Doesn’t work if any of the targets are 0.

R^2

Compares the **residual sum of squares (RSS)** to the **total sum of squares (TSS)**.

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$$TSS = \sum_{i=1}^{n} (\bar{y} - y_i)^2$$

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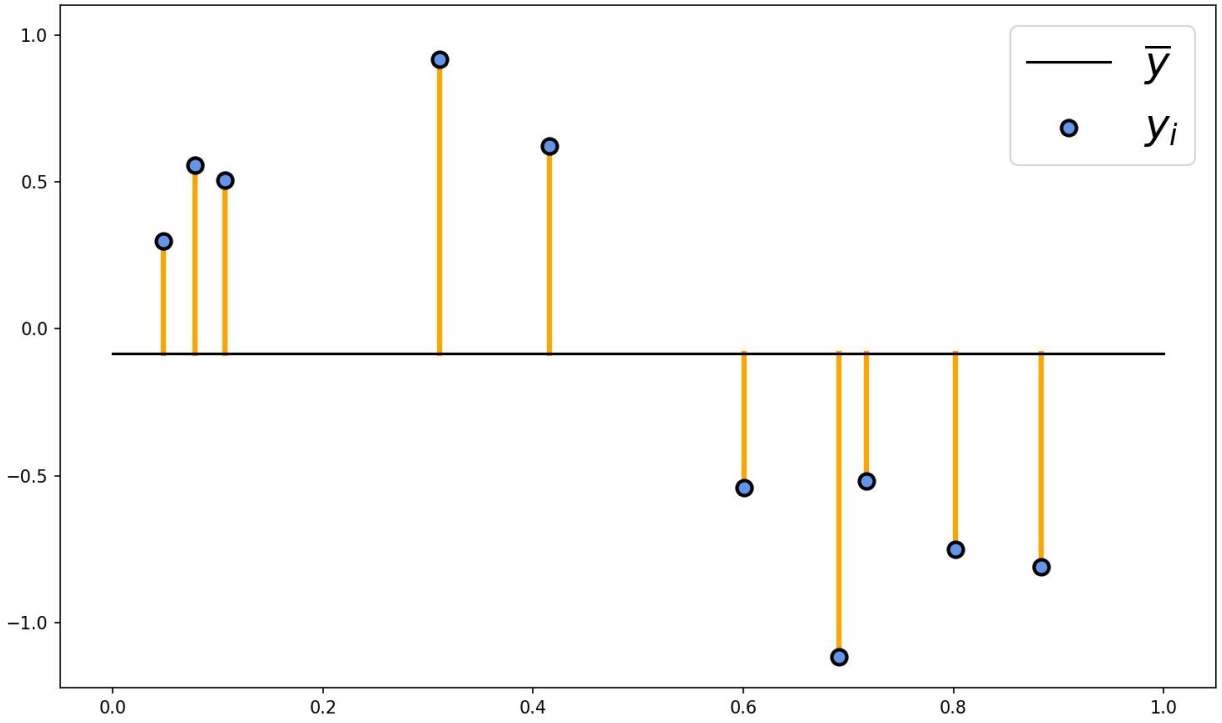
the average target value

Total Sum of Squares (TSS)



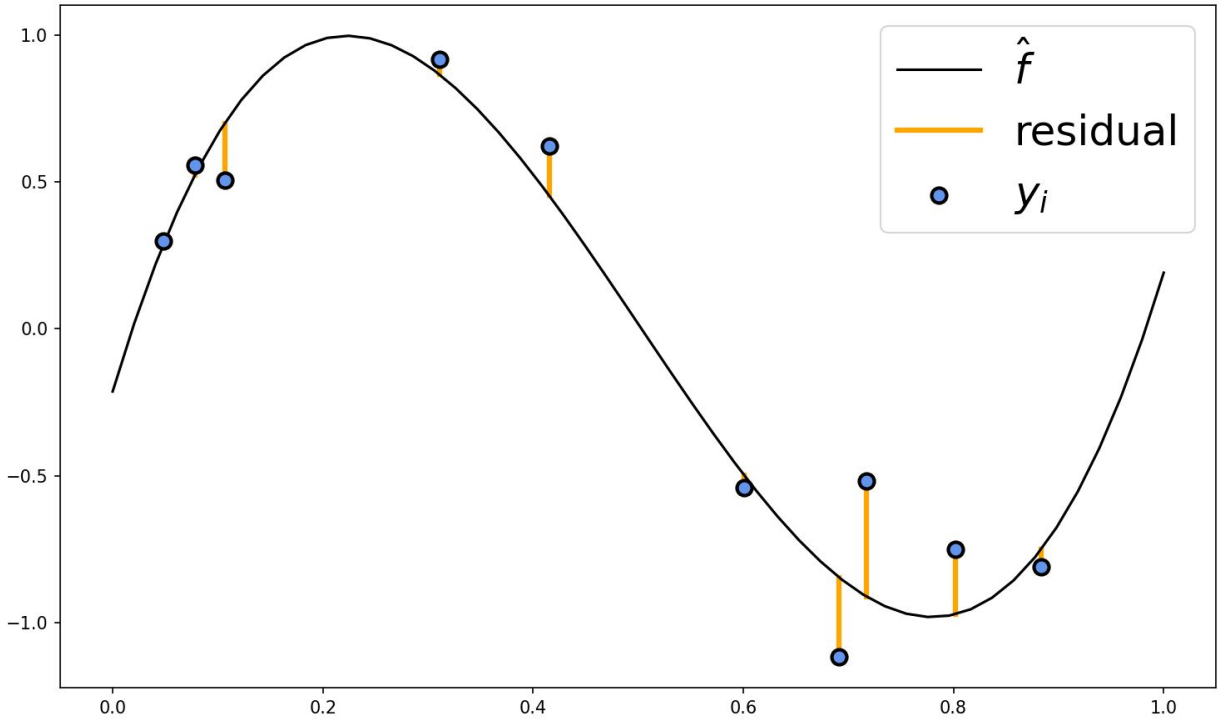
TSS = 4.845

R^2

Compares the **residual sum of squares (RSS)** to the **total sum of squares (TSS)**.

$$RSS = \sum_{i=1}^{n} (\hat{f}(x_i) - y_i)^2$$

Residual Sum of Squares (RSS)



RSS = 0.352

R^2

Compares the **residual sum of squares (RSS)** to the **total sum of squares (TSS)**.

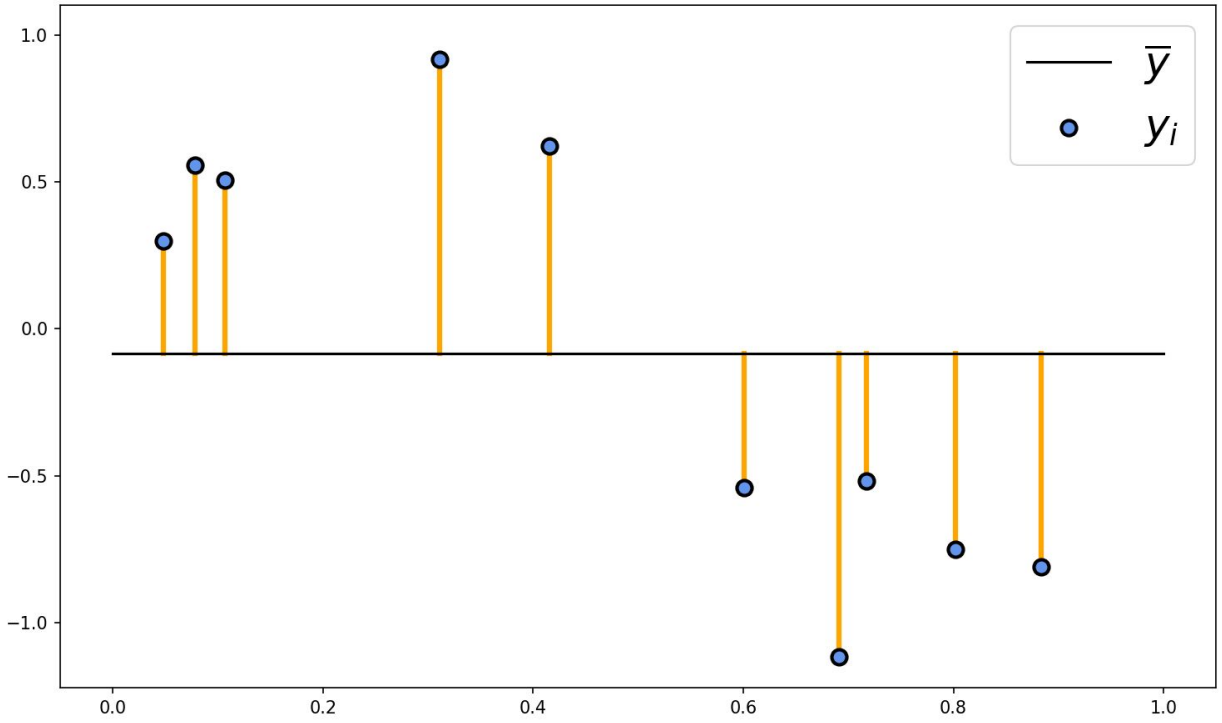
$$R^2 = \frac{TSS - RSS}{TSS}$$

The best possible R^2 of 1 occurs when RSS = 0.

The simplest model (predict the average target value) has an R^2 value of 0.

The range of R^2 is $(-\infty, 1]$.

Total Sum of Squares (TSS)



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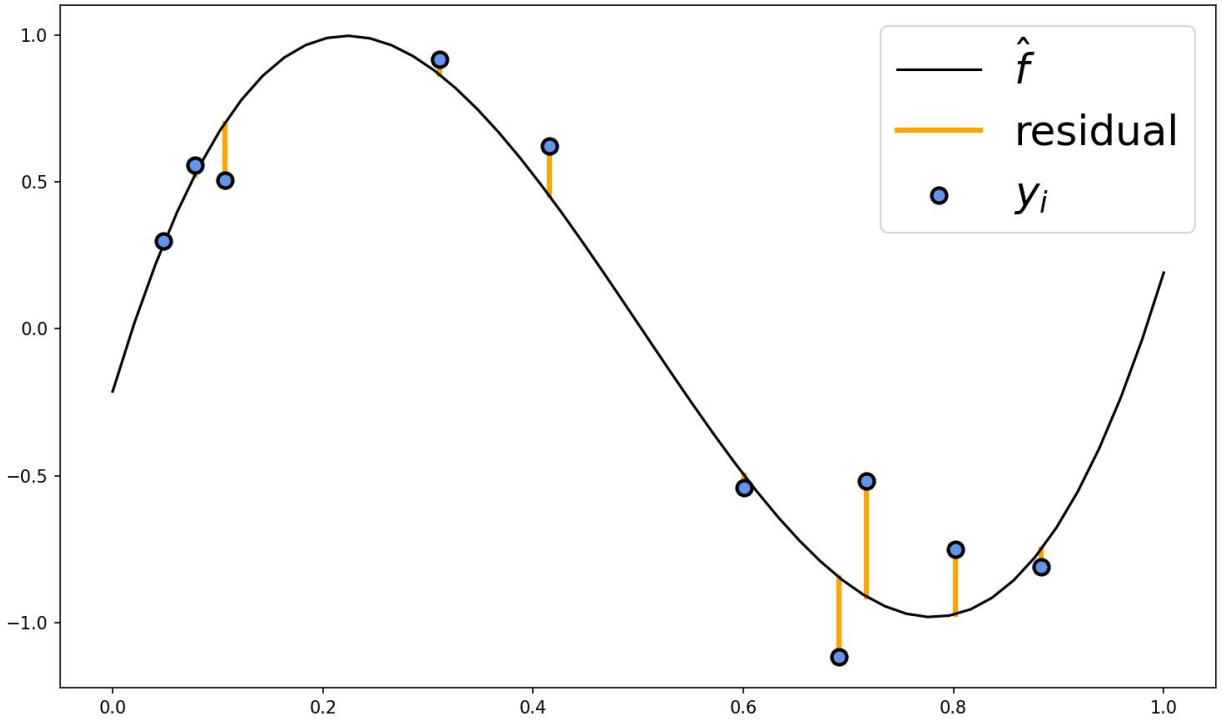
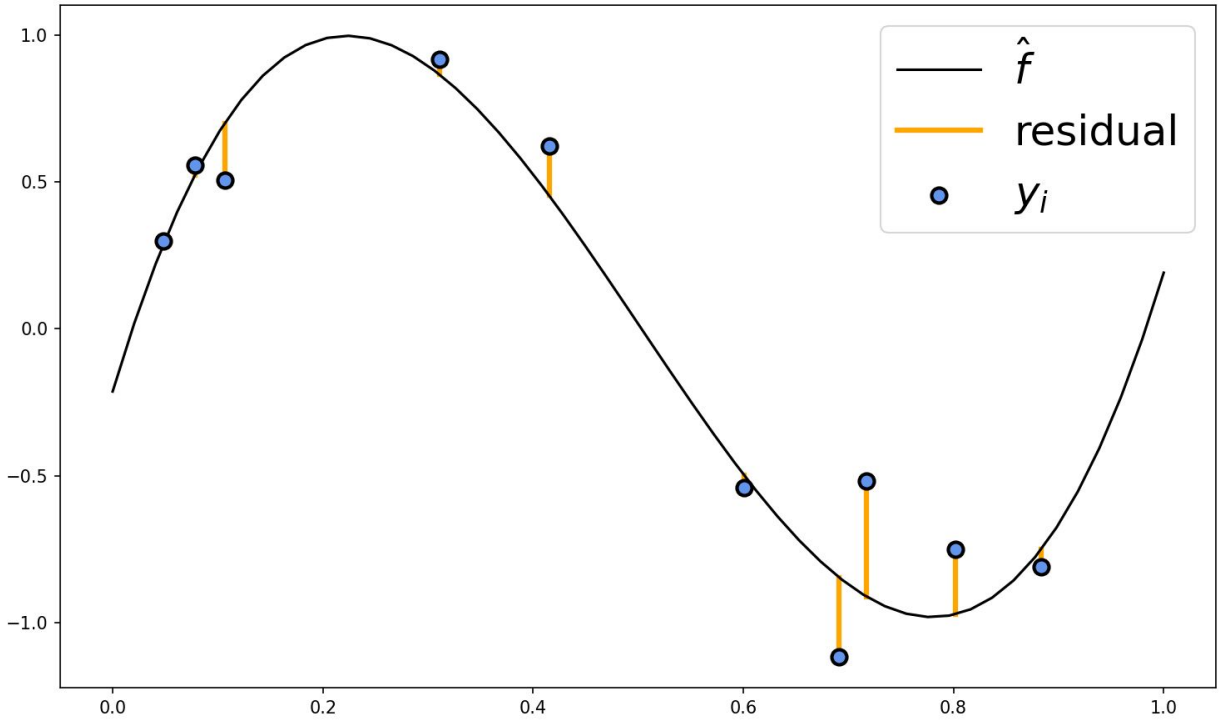


TSS = 4.845

RSS = 0.352

R^2



TSS = 4.845

RSS = 0.352

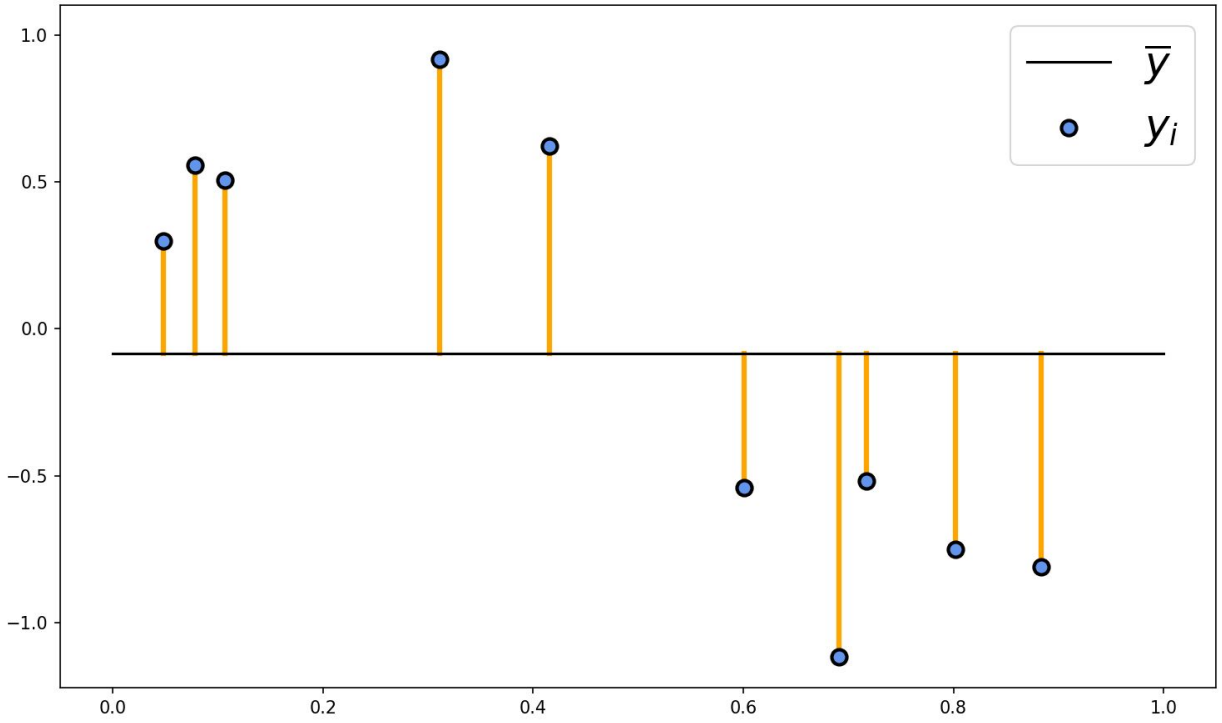
$$R^2 = \frac{4.845 - 0.352}{4.845}$$

$$= 0.931$$

R^2

What about a negative R^2 value?

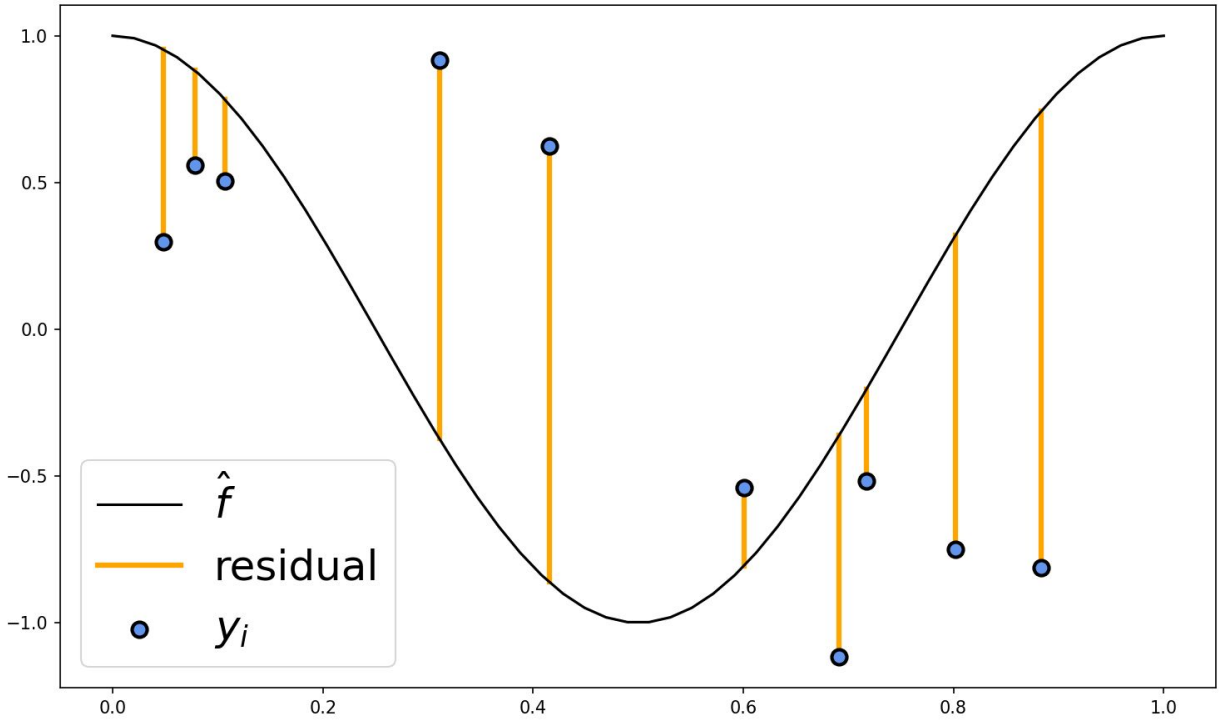
Total Sum of Squares (TSS)



TSS = 4.845

RSS = 0.352

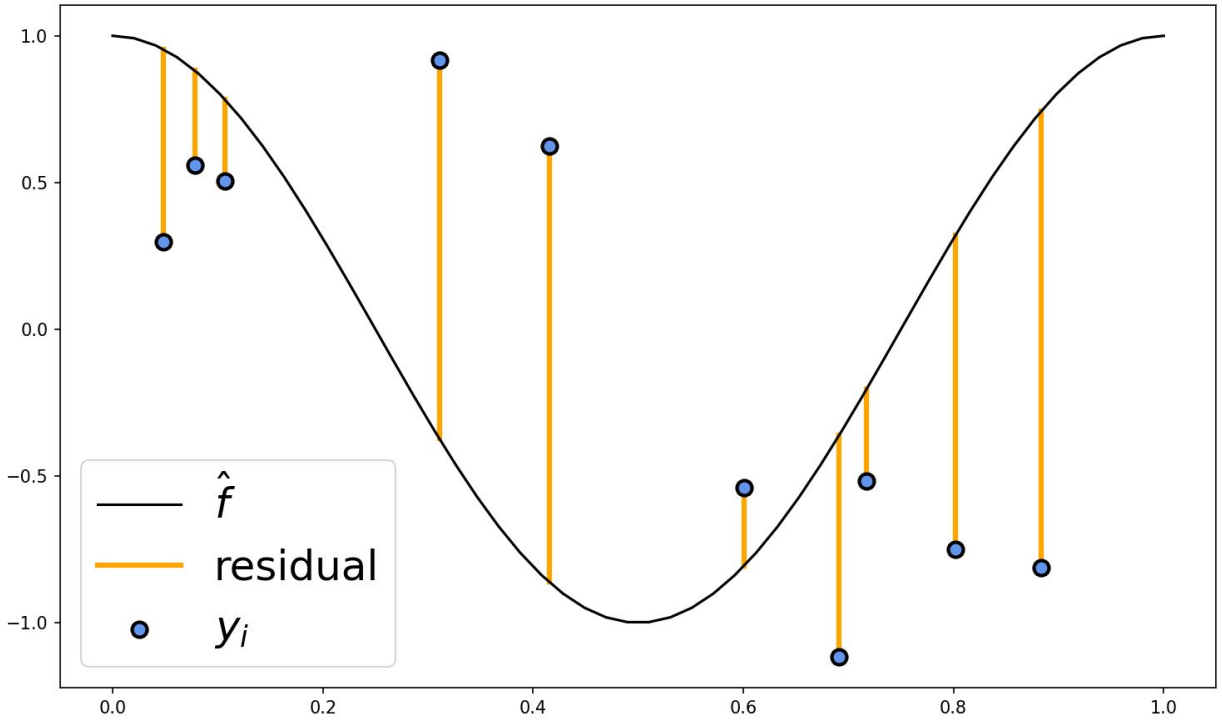
Residual Sum of Squares (RSS)



TSS = 4.845

RSS = 97.58

R^2



TSS = 4.845

RSS = 97.58

$$R^2 = \frac{4.845 - 97.58}{4.845}$$

$$= -19.140$$